

Project Title

ProManage AI – AI Powered Multi-Company Project Management and Collaboration Platform

1. Problem Statement

In modern organizations, managing multiple projects, teams, employees, and client communication efficiently is a major challenge. Many companies still rely on disconnected tools such as spreadsheets, emails, messaging platforms, and manual reporting systems. This creates several critical problems:

- Lack of centralized project management

- Managers cannot monitor all tasks, deadlines, and employee progress in one place.

- Poor real-time visibility

- Managers and company administrators cannot track task progress in real time, making it difficult to identify delays or productivity issues.

- Inefficient team collaboration

- Communication between managers, employees, and teams is fragmented across different platforms, leading to confusion and inefficiency.

- No intelligent performance analysis

- Most systems do not provide AI-based insights, automated performance reports, or meeting summaries.

- Lack of secure multi-company support

- Existing systems do not properly isolate company data, creating privacy and security risks.

- No integrated meeting and collaboration system

- Teams use external tools for meetings, file sharing, and communication, which reduces efficiency.

- Poor client visibility

- Clients cannot easily monitor project progress without exposing sensitive internal company data.

No automated reporting and productivity tracking

Managers manually generate reports, which is time-consuming and prone to errors.

Lack of scalable subscription-based platform

Companies need a SaaS-based platform with free trial and subscription model.

Therefore, there is a need for a centralized, secure, intelligent, real-time, AI-powered project management system that supports multiple companies, teams, and clients.

2. Proposed Solution

To solve the above problems, we propose to develop **ProManage AI**, a centralized AI-powered project management and collaboration platform built using the MERN stack, WebRTC, AI APIs, and Stripe subscription system.

The system provides a secure, scalable, multi-tenant SaaS platform where companies can manage projects, teams, tasks, meetings, communication, and performance analysis in one place.

The platform offers:

- Real-time task tracking
- AI-generated performance reports
- AI meeting summaries
- Secure role-based access control
- WebRTC video meetings
- Team collaboration and chat
- File and document sharing
- Client progress visibility
- Stripe subscription and billing system
- Multi-company secure architecture

This platform improves productivity, transparency, collaboration, and decision-making while ensuring security and scalability.

3. System Objectives

The main objectives of this system are:

- To provide centralized project management
- To enable real-time monitoring of tasks and employee performance
- To improve communication between managers, employees, and clients
- To automate report generation using AI
- To provide secure multi-company architecture
- To provide integrated meeting and collaboration tools
- To implement subscription-based SaaS platform
- To ensure data privacy and role-based access

4. User Roles in the System

The system supports multiple user roles with specific permissions.

4.1 Super Admin (Platform Owner)

This role manages the entire platform.

Responsibilities:

- Manage all companies
- View all subscriptions
- View payments and revenue
- Suspend or activate companies
- Monitor platform analytics
- Manage plans and pricing

Super Admin cannot access internal company task data.

4.2 Company Admin

This role represents the company owner.

Responsibilities:

- Create and manage departments
- Add managers and employees
- Manage company projects
- View company reports
- Manage company subscription
- Assign roles

Company Admin cannot access other company data.

4.3 Manager

Manager is responsible for managing projects and teams.

Responsibilities:

- Create projects
- Assign tasks to employees
- Set deadlines
- Monitor employee performance
- Schedule meetings
- Generate reports
- Communicate with team

Manager cannot access other teams or companies.

4.4 Employee

Employee performs assigned tasks.

Responsibilities:

- View assigned tasks
- Update task progress
- Submit daily work reports
- Join meetings
- Upload files
- Participate in team chat

Employee cannot access other projects or teams.

4.5 Client

Client monitors project progress.

Responsibilities:

- View project progress
- View deadlines
- View reports
- View meeting summaries

Client cannot access:

- Internal chat
- Employee performance data
- Internal files

5. System Features

5.1 Multi-Company Support (Multi-Tenant Architecture)

- Multiple companies can use the platform
- Each company's data is isolated
- Secure and private data access

5.2 Project Management

Managers can:

- Create projects
- Assign team members
- Set deadlines
- Track progress

5.3 Task Management

Managers can:

- Assign tasks
- Track completion status
- Monitor delays

Employees can:

- Update progress
- Submit work reports

5.4 Real-Time Monitoring

Using Socket.io:

- Live task updates
- Live notifications
- Online status tracking

5.5 AI Performance Analysis

AI automatically analyzes:

- Task completion
- Productivity
- Work logs

Generates:

- Performance reports
- Productivity insights

5.6 Meeting System (WebRTC)

Managers can:

- Schedule meetings
- Conduct video meetings

Employees can:

- Join meetings
- Collaborate in real time

5.7 AI Meeting Summary

AI automatically:

- Converts speech to text
- Generates meeting summary
- Stores summary in system

5.8 Chat System

Supports:

- Team chat
- Project chat
- Direct messaging

5.9 File Sharing System

Users can upload:

- PDFs
- Images
- Documents
- Code files

All files stored securely.

5.10 Notification System

Users receive alerts for:

- Task assignment
- Deadline reminder
- Meeting schedule
- File upload

5.11 Client Access Portal

Clients can view:

- Project progress
- Reports
- Meeting summaries

Without accessing internal company data.

5.12 Payment and Subscription System (Stripe)

Platform provides:

- Free trial period (5–6 months)
- Paid subscription plans
- Secure Stripe payment integration

After trial ends:

- Company must subscribe to continue

6. Admin Dashboard Features

Super Admin Dashboard:

- View all companies
- View payments
- Manage subscriptions
- Platform analytics

Company Admin Dashboard:

- Manage employees
- Manage managers

- Manage projects
- View reports

7. User Dashboard Features

Manager Dashboard:

- Create projects
- Assign tasks
- Monitor team
- Schedule meetings

Employee Dashboard:

- View tasks
- Submit reports
- Join meetings
- Upload files

Client Dashboard:

- View project progress
- View reports
- View summaries

8. Security and Privacy

The system provides:

- Role-based access control
- Company-level data isolation
- Secure authentication using JWT
- Encrypted password storage
- Protected APIs

9. Expected Outcome

The system will provide:

- Efficient project management
- Improved productivity
- Real-time monitoring
- AI-based insights
- Secure collaboration
- Scalable SaaS platform

10. Conclusion

ProManage AI provides a complete, intelligent, secure, and scalable solution for modern project management. It integrates project tracking, AI analysis, communication, meetings, file sharing, and subscription management into a single platform.

This system improves efficiency, transparency, and collaboration while ensuring data privacy and security.